SECTION 7.C
Half-Angle, Sum, and Difference Identities

Learning Objectives

- I can apply half-angle formulas for sine and cosine to evaluate exact expressions.
- I can apply half-angle formulas for tangent and determine appropriate signs.
- I can apply angle sum and difference identities to rewrite and evaluate trigonometric expressions.
- I can justify multi-step reasoning using algebraic structure.

Introduction

In Section 7.B, we rewrote trigonometric expressions involving **double angles**.

Now we extend that structure.

Instead of working with 2θ, we will:

- Break angles apart
- Combine angles strategically
- Rewrite unfamiliar angles using known unit circle values

Structure matters.

Key Vocabulary

Half-Angle Formula: An identity that expresses a trig function of $\frac{\theta}{2}$ in terms of θ.

Sum Identity: A formula that expresses a trig function of $A + B$.

Difference Identity: A formula that expresses a trig function of $A - B$.

Sign Determination: Deciding whether a half-angle expression is positive or negative based on quadrant.

Core Identities

Half-Angle

$\sin\left(\frac{\theta}{2}\right) = \pm\sqrt{\frac{1-\cos\theta}{2}}$

$\cos\left(\frac{\theta}{2}\right) = \pm\sqrt{\frac{1+\cos\theta}{2}}$

$\tan\left(\frac{\theta}{2}\right) = \frac{\sin\theta}{1+\cos\theta}$

Sum and Difference

$\sin(A+B) = \sin A\cos B + \cos A\sin B$

$\sin(A-B) = \sin A\cos B - \cos A\sin B$

$\cos(A+B) = \cos A\cos B - \sin A\sin B$

$\cos(A-B) = \cos A\cos B + \sin A\sin B$

Structure Insight

- Half-angle formulas contain square roots, introducing ±.
- The sign depends on the quadrant of $\frac{\theta}{2}$.
- Cosine switches signs between sum and difference forms.
- Precision in signs is critical.

Examples

Example 1: Evaluate $\sin\left(\frac{\pi}{8}\right)$.

Example 2: Evaluate $\cos(75°)$.

Example 3: Rewrite $\sin(45° + 30°)$.

Example 4: Simplify $\cos(A - B) + \cos(A + B)$.

Summary

- Half-angle identities introduce square roots and ± signs.
- Signs depend on quadrant location.
- Sum and difference identities rewrite unfamiliar angles.
- Structure and precision are essential.

Common Mistakes

- Forgetting the ± in half-angle formulas.
- Using the wrong quadrant for sign.
- Mixing up cosine sign patterns.
- Not fully simplifying radicals.

Practice Problems

Easy

1. $\sin\left(\frac{\pi}{4}\right)$

2. $\cos\left(\frac{\pi}{6}\right)$

3. Rewrite $\sin(45° + 30°)$.

4. Rewrite $\cos(45° - 30°)$.

Medium

5. Evaluate $\sin(75°)$.

6. Evaluate $\cos(15°)$.

7. Simplify $\tan\left(\frac{\pi}{8}\right)$ if $\cos\theta = \frac{1}{3}$.

8. Expand $\sin(A - B)$.

Harder

9. Evaluate $\sin\left(\frac{\pi}{12}\right)$.

10. Evaluate $\cos\left(\frac{\pi}{12}\right)$.

11. Verify $\sin(2A) = 2\sin A\cos A$.

12. Simplify $\cos(A + B) + \cos(A - B)$.

Challenge

13. Express $\tan\left(\frac{\theta}{2}\right)$ in terms of cosine only.

14. Evaluate $\sin\left(\frac{\pi}{24}\right)$.

15. Simplify $\cos(75°) + \sin(15°)$.

16. Rewrite $\cos(2A)$ using a sum identity only.

Justify

17. Explain why half-angle identities require ±.

18. Justify how sum identities extend unit circle reasoning.

Solutions

Easy

1. $\frac{\sqrt{2}}{2}$

2. $\frac{\sqrt{3}}{2}$

3. $\frac{\sqrt{6}+\sqrt{2}}{4}$

4. $\frac{\sqrt{6}+\sqrt{2}}{4}$

Medium

5. $\frac{\sqrt{6}+\sqrt{2}}{4}$

6. $\frac{\sqrt{6}+\sqrt{2}}{4}$

7. Simplified algebraic expression

8. $\sin A\cos B - \cos A\sin B$

Harder

9. $\frac{\sqrt{6}-\sqrt{2}}{4}$

10. $\frac{\sqrt{6}+\sqrt{2}}{4}$

11. Identity verified

12. $2\cos A\cos B$

Challenge

13. Algebraic form

14. Exact radical form

15. Simplified radical expression

16. $\cos^2 A - \sin^2 A$

Justify

17. Square roots produce both positive and negative results; quadrant determines sign.

18. They break complex angles into familiar unit circle angles.